SIGNATURES

BPDRPT AZD8205 Purification Process Description

Contributors to the preparation of this document
Preparer(s) [redacted]
Approver(s) [redacted]

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1 INTRODUCTION

1.1 Purpose

This report describes the conjugation process and process parameters for the production of AZD8205 Drug Substance. The overall process is designated “Process 1”.

Table 1.1 General Properties: AZD8205 Drug Substance

Property	Description
Product classification	Antibody Drug Conjugate
Payload	SG3932
Target DAR	8.0
Isoelectric point (pI) range	8.2-9.0
Extinction coefficient mAb Conjugated ADC	mAb: 1.58 ADC: 1.98
mAb Intermediate molecular weight (g/mol)	144,695
Payload molecular weight (g/mol)	1148.3
mAb Intermediate concentration (g/L)	50 (range: 45-55)
Overall expected conjugation yield (%)	90%
Formulated Drug Substance concentration (g/L)	20.0
Drug Substance formulation	20 mM histidine/histidine-HCl, 240 mM sucrose, 0.02% (w/v) polysorbate 80, pH 6.0
Drug substance presentation	Stored frozen at ≤ - 65°C in HDPE bottles in light protecting bag.

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1.2 Assumptions

- This report does not consider scale or facility constraints.
- Unless otherwise noted, procedures pre/post common unit operations should be followed for filters and unit operations per GUID-0024075.
- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- The target volumes are expected to be sufficient. Overage to account for system hold up volume is site dependent and should be a separate phase and not added into the volume targets for the process.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (15-27°C).
- For facilities for which 1.0 N NaOH is not feasible, 0.5 N NaOH may be used as a substitute with "Conductivity (mS/cm @ 25°C), final" NOR: 84 – 113.
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Unless otherwise noted, all process parameters are inputs.
- Safety practices and procedures defined by SHE should be followed at all times and can supersede details described in this document.

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1.3Abbreviations/Acronyms/Definition

Table 1.2 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
AEX	Anion exchange chromatography	LMH	Liters/meter^2-hour
ADC	Antibody Drug Conjugate	MV	Membrane volume
AL	Alert limit	NA	Not applicable
Bldg	Building	NC	Non-conformance
BR	Batch record	NOR	Normal operating range
CEX	Cation exchange chromatography	OD	Optical density
Conc.	Concentration	PCR	Polymerase chain reaction
CV	Column volume	PES	Polyethersulfone
DAR	Drug Antibody Ratio	PFB	Pre-formulated bulk
DBC	Dynamic binding capacity	P/N	Part number
DS	Drug substance	PP	Polypropylene
DV	Diafiltration volumes	psi; psig	pounds per square inch; pounds per square inch gauge
EAB	Excipient addition buffer	qs	Bring to final weight (quantum satis)
ELISA	Enzyme-linked immunosorbent assay	RPHPLC	Reverse phase high performance chromatography
FB	Formulated bulk	S.F.	Safety factor
FIO	For information only	SS	Stainless steel
HDPE	High density polyethylene	TBD	To be determined
HETP	Height equivalent to a theoretical plate	TMP	Trans-membrane pressure
HPLC	High-performance liquid chromatography	UDS	Unformulated drug substance
HPSEC	High-performance size-exclusion chromatography	UF/DF	Ultrafiltration/diafiltration
HCP	Host cell protein	WFI	Water for injection
IEC	Ion exchange chromatography		
LG	Linear gradient		
LR	Lot release		

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1.4Definitions of Terminologies for Parameter Ranges

1.2.1 Target

The target value is a preset value that a process parameter is aimed to by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Alert Limit or Normal Operating Range.

1.2.2 Normal Operating Range (NOR)

Operating parameter ranges, also called Action Limits, which are established based on current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value exceeding a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

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2 BILL OF MATERIALS

Table 2.1

Type	Component Name	Minimum Grade	Part Number (s)	Justification/Notes
Product Intermediates	INT-016	NA	NA	NA
	Payload SG3932	NA	NA	NA
Pre-packed Chromatography media	Ceramic Hydroxyapatite (CHT) Type I, 40µm	NA	NA	NA
Buffer and solution components	Ethylenediamine-tetraacetic acid, disodium salt, dihydrate (EDTA)	USP	4100799 6003688	NA
	Dimethyl Sulfoxide (DMSO)	USP	4102623 6000316 6001867	NA
	L-Histidine HCl monohydrate	JP, EP	4101011	NA
	L-Histidine	EP, JP	4101095	NA
	Polysorbate 80	NF, EP, JP, USP	4100792	NA
	N-acetyl Cysteine (NAC)	GMP Manufactured	6003593	NA
	Sodium Chloride	USP	6000324	NA
	10N Sodium hydroxide	GMP Manufactured	6002721	NA
	Sodium hydroxide, pellets	BP, Ph. Eur., JP. NF	4101097	NA
	Sodium phosphate, dibasic, hepta (7) hydrate	USP	6002094 6001964 4101097	NA
	Sodium phosphate, monobasic, monohydrate	BP, USP	4101091 6002047 6002096	NA
	Sucrose (low endotoxin)	USP, BP, JP	4102624	NA
	Tris base (tromethamine/trometamol)	USP, EP	4101399	NA
	Tris(2-carboxyethyl) phosphine (TCEP) HCl	GMP	6004537 6004538	NA
Filters	Pellicon 3 Ultracell capsules C Screen 30kDa MWCO	NA	6004003 6004004	0.1 m^2 0.5 m^2 1.1 m^2
	Durapore Opticap 0.2 µm	NA	NA	Sterile Grade 0.2 µm

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Type	Component Name	Minimum Grade	Part Number (s)	Justification/Notes
Miscellaneous	HDPE Bottles	NA	4102052 4102055	1 L 4 L

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3 IN-PROCESS HOLD TIMES

Hold times begin when product is generated and ends when the next step is completed.

Unless otherwise noted, room temperature hold times are additive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 2 days at room temperature prior to being stored 1 day refrigerated would have an additive hold time of 3 days at refrigerated temperature but only 2 days at room temperature.

Table 3.1 Recommended Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time NOR-Max (hours, 15-25 °C)	Refrigerated Temperature Hold Time NOR-Max (hours, 2-8 °C)	Justification/Notes
mAb Intermediate	≤ 7 days	≤ 60 days	NA
Diluted mAb Intermediate	≤ 24	≤ 24	NA
Quenched Intermediate	≤ 24	≤ 24	Product must be protected from light.
CHT Product	≤ 24	≤ 24	Product must be protected from light.
Pre-formulated Intermediate	≤ 24	≤ 24	Product must be protected from light.
Formulated Bulk	≤ 72	≤ 96	Product must be protected from light.

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4 LOAD AND PRODUCT FILTRATION

Table 4.1 Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m^2)	Justification/ Notes
Diluted mAb Intermediate	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	115	Volumetric load NOR MAX based on achieved challenge at-scale of ~114 L/m^2.
CHT Chromatography (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	TBD	Achieved challenge at scale ~35L/m^2.
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	TBD	Achieved challenge at scale ~181 L/m^2.
Concentration and Diafiltration (LF (optional) → Unit Op →PF)	Load Filter (optional)	Express SHC, 0.5/0.2 µm	Fed batch	580	Volumetric load NOR MAX based on achieved challenge at scale ~577 L/m^2.
	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	70	Volumetric load NOR MAX based on achieved challenge at scale ~70 L/m^2.
Conjugated Drug Substance Formulation and Bulk Filtration (Unit Op → Bulk filter	Bulk Filtration	Durapore Opticap, 0.2 µm	Off-line	245	Volumetric load NOR MAX based on achieved challenge at scale ~243 L/m^2.

Millipore SHC filters may be interchanged with Pall Fluorodyne EX EDF filters of the same device size. This assumes the additional filter area of the Pall device compensates for any difference in maximum volumetric load.

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5 MAB PREPARATION

Table 5.1 mAb Preparation: Process Parameters

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Adjustment to 20g/L protein concentration *20mM Sodium Phosphate, 1mM EDTA, pH 7.0*	Conc, (g/L)	20.0	18.0	22.0	~1.4 kgs of buffer to kg mAb intermediate
pH Adjustment *1M Tris base*	pH	7.0	6.8	7.2	~0.008 kg base/ kg mAb intermediate The starting amount of mAb intermediate should be adjusted based on the amount of payload weighed. Density of diluted mAb intermediate = 1.016 g/mL

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6 MAB REDUCTION

Table 6.1 mAb Reduction: Process Parameters

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Temperature Control	°C	20.0	18.0	22.0	NA
Reduction *50mM Neutralized TCEP*	Mole TCEP to mole mAb intermediate	10.0	9.5	10.5	~0.064 kg TCEP solution to kg mAb intermediate
	Time (mins)	120	110	180	Gentle mixing without foaming and entrainment of air. Incubation time ends when the volume addition of the next step is fully completed.

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7 MAB CONJUGATION

Table 7.1 mAb Conjugation: Process Parameters

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Temperature Control during payload addition and incubation	°C	21	18.0	28.0	Exothermic reaction. Ensure temperature control is adequate to avoid exceeding NOR max. Control rate of payload addition to help mitigate temperature increase.
Conjugation *10mM payload in DMSO*	Mole payload to mole mAb intermediate	14.0	13.5	14.5	Target 14x molar ratio. Payload is dissolved in DMSO solution when delivered to the reduced mAb intermediate. Assume a density of 1.1g/mL for payload stock.
	Payload in DMSO addition time (mins)	NA	NA	40	NA
	Incubation time (mins)	180	170	190	Gentle mixing without foaming and entrainment of air. Incubation time ends when the volume addition of the next step is fully completed.
Quenching *100mM NAC*	Mole NAC to mole payload added	4.0	3.5	4.5	Target 4x molar ratio of NAC to payload assume density of 1 mg/mL Density of crude conjugate is 1.015g/mL
	Time (mins)	60	55	720	Gentle mixing without foaming and entrainment of air Quench time ends when full volume loaded onto CHT column
	Temperature Control during incubation (°C)	19.0	18.0	28.0	NA

NOTE: KEEP LOAD PRODUCT AND COLLECTION VESSEL COVERED FROM LIGHT AS MUCH AS POSSIBLE DURING THE PROCESS, WITHOUT THE CREATING A HINDERANCE TO NORMAL OPERATION. COVER SAMPLES IMMEDIATELY AFTER ALIQUOTING.

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8 CHT CHROMATOGRAPHY

Table 8.1 CHT Chromatography: Process Parameters

Step	Parameter (units)	Target	NOR		Justifications/notes
			Min	Max	
General	Media	CHT, type 1, 40 um			Prep per appropriate SOP
	Bed height (cm)	20	18	22	NA
Pre-Equilibration *300 mM sodium phosphate, pH 7.0*	Linear Velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	3.0	2.5	NA	NA
Equilibration 1 *10 mM sodium phosphate, pH 7.0*	Linear velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	3.0	2.5	NA	NA
	pH [a]	NA	6.8	7.2	Run and record
	Conductivity (mS/cm)	NA	NA	NA	Run and record
Load Quenched Conjugated product	Linear velocity (cm/hr)	150	NA	220	NA
	Protein load (g / L resin)	NA	10	45	Max is based on binding capacity.
Equilibration 2 *10 mM sodium phosphate, pH 7.0*	Linear velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	5.0	4.5	NA	NA
Elution *10 mM sodium phosphate, 350 mM NaCl, pH 7.0*	Linear velocity (cm/hr)	200	NA	220	NA
	Start collection OD_{280}	1.0	0.8	NA	Volume before pre-peak = ~ 2.5 CV Volume before main peak = ~ 4.6 CV
	End collection OD_{280}	0.5	0.3	NA	Product volume = ~5 -7 CV

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Step	Parameter (units)	Target	NOR		Justifications/notes
			Min	Max	
Strip 1 *10 mM sodium phosphate, 1 M NaCl, pH 7.0*	Linear velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	5.0	4.5	NA	NA
Strip 2 *300 mM sodium phosphate, pH 7.0*	Linear Velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	3.0	2.5	NA	NA
Caustic Strip *0.5 N sodium hydroxide, 5 mM sodium phosphate*	Linear Velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	3.0	2.5	NA	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm) [a]	NA	84	113	NA
End of Cycle					
Sanitization Hold	Contact time (minutes)	35	15	120	NA
Storage *0.1 N sodium hydroxide, 5 mM sodium phosphate*	Linear Velocity (cm/hr)	200	NA	220	NA
	Volume (CV)	3.0	2.5	NA	NA
	Conductivity, effluent, (endpoint check) (mS/cm) [a]	NA	19	27	NA

[a] Output parameter

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9 CONCENTRATION AND DIAFILTRATION

Table 9.1 Concentration and Diafiltration: process Parameters

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
General	**Media**	**Pellicon 3 Ultracell Capsules C Screen 30kDa MWCO**			**Prep per appropriate SOP**
	Feed pressure (psig)	**NA**	**NA**	**60**	**NA**
Equilibration *Diafiltration buffer:* *20mM histidine,* *pH 6.0*	pH [a]	6.0	5.8	6.2	NA
Concentration 1 Control Method: TMP and Feed Flow Control	Membrane protein load (g/m^2)	NA	NA	300	(Mass of protein)/(area of ultrafiltration membrane)
	TMP (psig)	20	NA	25	NA
	Feed flow (L/min/m^2)	5.0	3.0	7.0	NA
	Conc. 1, final (g/L)	15	10	20	May be directly measured or calculated from permeate or retentate volume
Diafiltration *Diafiltration buffer:* *20mM histidine,* *pH 6.0* Control Method: TMP and Feed Flow Control	TMP (psig)	20	NA	25	NA
	Feed flow (L/min/m^2)	5.0	3.0	7.0	NA
	Diafiltration volumes	6.5	6.0	NA	(kg buffer)/(kg product)
	Permeate pH, final [a]	6.0	5.8	6.2	NA
	Retentate pH, final [a]	6.0	5.8	6.2	NA

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Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Concentration 2 Control Method: TMP and Feed Flow Control	TMP (psig)	20	NA	25	NA
	Feed flow (L/min/m^2)	NA	NA	7.0	NA
	Conc. 2, final (g/L)	35	25.4	40	May be directly measured or calculated from permeate or retentate volume
Recirculation	Feed pressure (psig)	20	NA	≤ 60	NA
	Feed flow (L/min/m^2)	NA	NA	7.0	NA
	Recirculation time (min)	5	NA	120	NA
Recovery Flush *Diafiltration buffer:* *20mM histidine,* *pH 6.0* Recovery Method: Buffer flush of system hold up volume	Feed pressure (psi)	20	NA	30	NA
	Feed flow (L/min/m^2)	NA	NA	7.0	NA
	Recovery conc. (g/L)	30	25.4	40	Recovery product density = 1.005 g/mL
Dilution to Pre-formulated intermediate *Diafiltration buffer:* *20mM histidine,* *pH 6.0*	PFI conc. (g/L)	26.7	25.4	28	NA
	pH [a]	6.0	5.8	6.2	NA

[a] Output parameter

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10 FORMULATION AND BULK FILTRATION

Table 10.1 Formulation and Bulk filtration: Process Parameters

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
EAB spike (of PFI to UB): *EAB: 20 mM histidine, 960 mM sucrose, pH 6.0* (density 1.127 g/mL)	Ratio of EAB mass to PFI mass (kg/kg)	0.376	0.367	0.385	UDS: *20.0 g/L AZD8205 in 20mM Histidine, 240 mM sucrose, pH 6.0* (measured density 1.038 g/mL)
	Protein concentration, endpoint check (g/L)	20.0	NA	NA	NA
	Osmolality (mOsm/kg)[a]	316	253	379	NA
FSS Spike (of UB to FB) *EAB: 10% PS-80* (density 1.01 g/mL)	Ratio of FSS mass to UB (g/kg)	2.00	1.95	2.05	FB/DS: *20.0 g/L AZD8205 in 20mM Histidine, 240 mM sucrose, 0.02% PS 80 (w/v), pH 6.0* (measured density 1.038 g/mL)
0.2µm Bulk Filtration of FB to DS	Capacity (L/m^2)	NA	NA	245	Challenge achieved at-scale 243L/m^2
	Drug substance container	HDPE bottle			DS filled HDPE bottles need to be placed in bag to be protected from light.
	Fill volume per container (%)	70	NA	90	If multiple containers are required, target equivalent volume in each.
	Drug substance storage temperature (°C)	-80	NA	-70	Drug substance can be held 7 days at 2-8°C or 3 days at 15-25°C prior to freeze.
Integrity Test *0.2 µm Filter*	Integrity Test Result (PASS/FAIL)[a]	PASS	PASS	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

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11 BUFFERS AND SOLUTIONS

Buffer and solution stability studies are a commercial process definition activity. However, buffers and solutions with completed studies because of prior experience or a stability concern have stated hold times below. The studies were performed by holding buffers and solutions at ambient temperature in stainless steel containers. The studies were designed to test chemical stability for longer than the expected maximum time for routine storage. These times may be extended by an appropriate study unless otherwise noted. The studies do not address microbial control, which should be defined by the facility. Assays used to measure stability included pH and conductivity. In some cases, the hold time is measured directly (direct). In other cases, the hold time is supported by the hold time of similar buffers if the proposed buffer falls within our current buffer matrix (matrix). Please see REP-0202700 for reference.

Table 11.1 Buffer and Solution References

Buffer Name	Recipe Table	Doc Num/ AZ PN	Hold Time-Max (days, 15-25°C)	Unit Operation	Steps
20mM Sodium Phosphate, 1mM EDTA, pH 7.0	PRO-0188988	MABR-0011701/ 8009044	30 (matrix)	MAB Preparation	Concentration Adjustment
1 M Tris base	PRO-0188988	A0101-D	30 (matrix)	MAB Preparation	pH Adjustment
50 mM Neutralized TCEP	PRO-0188988	MABR-0011646/ 8009045	24 hours	MAB Reduction	Reduction
10 mM payload in DMSO	PRO-0188988	8009046	24 hours	MAB conjugation	Conjugation
100 mM NAC	PRO-0188988	MABR-0011689/ 8009047	24 hours	MAB conjugation	Quenching
300 mM sodium phosphate, pH 7.0	PRO-0188988	A-0446	72 (matrix)	CHT chromatography	Pre-Equilibration Strip 2

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Buffer Name	Recipe Table	Doc Num/ AZ PN	Hold Time-Max (days, 15-25°C)	Unit Operation	Steps
10 mM sodium phosphate, pH 7.0	PRO-0188988	A-005297/ 8002901	30 (matrix)	CHT chromatography	Equilibration 1
					Equilibration 2
10 mM sodium phosphate, 350 mM NaCl, pH 7.0	PRO-0188988	MABR-0011691/ 8009048	30 (matrix)	CHT chromatography	Elution
10 mM sodium phosphate, 1 M NaCl, pH 7.0	PRO-0188988	8010243	30 (matrix)	CHT chromatography	Strip 1
0.5 N sodium Hydroxide, 5mM sodium phosphate	PRO-0188988	MABR-0011685/ 8009049	30 (matrix)	CHT chromatography	Caustic Strip
0.1 N sodium hydroxide, 5 mM sodium phosphate	PRO-0188988	Azdoc0213731/ 8009050	30 (matrix)	CHT chromatography	Storage
20 mM histidine/histidine-HCl, pH 6.0	PRO-0188988	A-0643	30 (matrix)	Concentration and Diafiltration	Equilibration
					Diafiltration
					Recovery Flush
					Dilution
20 mM histidine/histidine-HCl, 960 mM sucrose pH 6.0	PRO-0188988	A-005465/ 8005513	25	Formulation and Bulk Filtration	EAB spike
10% PS-80	PRO-0188988	A-0647	30 (direct)	Formulation and Bulk Filtration	FSS spike

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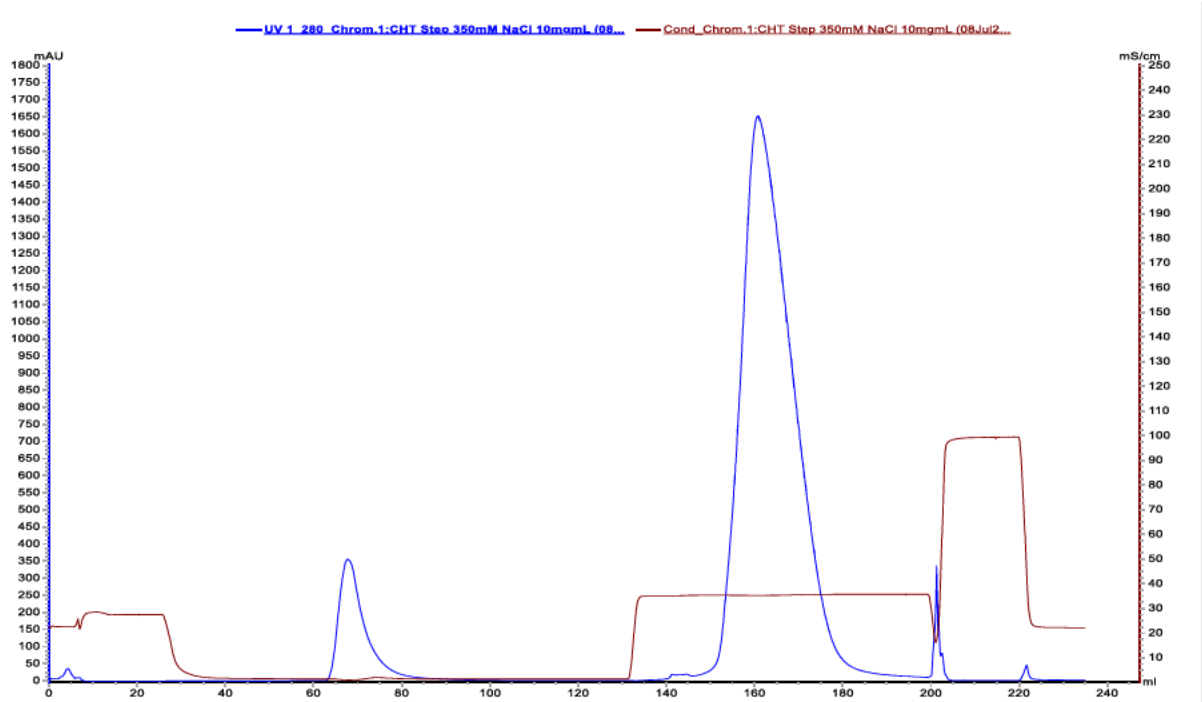
12 REFERENCE CHROMATOGRAMS

Sample chromatograms representing the chromatography step used in the purification process are presented below.

Table 12.1 CHT Chromatogram

Resin:	CHT Type I 40µm
Column Dimensions:	30 cm (D) x 20 cm (H) - 14.1 L (CV)
Notes:	NA

Figure 12-1 CHT Chromatogram



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13 PROCESS CHANGES

Table 13.1 lists changes in the AZD8205 Conjugation Process Description from edition 001 to edition 002. These changes do not warrant a process change from Process 1.

Table 13.1 Process Changes

Unit Operation	Step	Parameter	Original	New	Summary/Justification
mAb reduction	Reduction	Time (mins)	**ED001:** NOR max = 130 mins	**ED002:** NOR max = 180 mins	Increased for manufacturing flexibility.
	Reduction	Mole TCEP to mole mAb intermediate	**ED001:** NOR = N/A	**ED002:** NOR = 9.5 – 10.5	Defined ranges for manufacturing.
mAb conjugation	Conjugation	Mole payload to mole mAb intermediate	**ED001:** NOR = N/A	**ED002:** NOR = 13.5 – 14.5	Defined ranges for manufacturing.
	Conjugation	Payload in DMSO addition time (mins)	**ED 3.0: NA**	**ED 4.0 : NOR= 40**	Added a new parameter for payload in DMSO addition time to control payload addition.
	Quenching	Mole NAC to mole payload added	**ED001:** NOR = N/A	**ED002:** NOR = 3.5 – 4.5	Defined ranges for manufacturing.

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	Quenching	Time (mins)	**ED001:** NOR = 65	**ED002:** NOR = 720	Increased for manufacturing flexibility.
CHT chromatography	Equilibration 1	pH	**ED001:** NOR = N/A	**ED002:** NOR = 6.8 – 7.2	Defined ranges for manufacturing.
	Caustic Strip	Conductivity, effluent (endpoint check) (mS/cm)	**ED001:** NOR = N/A	**ED002:** NOR = 84 – 113	Defined ranges for manufacturing.
	Storage	Conductivity, effluent (endpoint check) (mS/cm)	**ED001:** NOR = N/A	**ED002:** NOR = 19 – 27	Defined ranges for manufacturing.
	Elution	Start collection OD	**ED001:** 0.2	**ED002: 1**.0	Changed to avoid collecting pre-peak.
	Strip 1	Buffer	**ED001:** 300mM sodium phosphate, pH 7.0	**ED002:** 10mM sodium phosphate, 1M NaCl, pH 7.0	Used in cleanability study and adopted for GMP manufacturing
	Strip 2	Buffer	**ED001:** N/A	**ED002:** 300mM sodium phosphate, pH 7.0	Same buffer used in a different step.

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	Load	Protein load (g/L resin)	**ED 3.0** NOR max = 30	**ED 4.0** NOR max = 45	Increased NOR max for column protein load. Supported by PPS data.
Concentration and Diafiltration	Concentration 2	Conc. 2, final (g/L)	**ED001:** Target = 30.0 AL = 27 – 33 NOR ≥ 25.4	**ED002:** Target = 35 AL = 27 – 39 NOR = 25.4 – 40	Increased target and widened AL range for better manufacturing flexibility. Added NOR max to avoid over-concentration.
	Recovery Flush	Recovery conc. (g/L)	**ED001:** Target = NA NOR ≥ 25.4	**ED002:** Target = 30 NOR = 25.4 – 40	Defined target for manufacturing. Added NOR max to avoid over-concentration.
	Concentration 1	Membrane protein load (g/m^2)	**ED001:** Target = 100 AL max = 200 NOR max = 400	**ED002:** Target = N/A AL max = 100 NOR max = 300	Initial ranges were set based on facility fit.
Formulation and Bulk Filtration	EAB spike (of PFI to UB)	Osmolality (mOsm/kg)	**ED001:** Target = TBD AL = TBD	**ED002:** Target = 314 AL = 267 – 361	Defined target and range for manufacturing.
	EAB spike (of PFI to UB)	Osmolality (mOsm/kg)	**ED002 :** Target = 315 AL = 267-361 NOR= NA	**ED 3.00:** Target = 316 AL= NA NOR= 253-379	Updated target value based on DFDD calculated theoretical value. Removed the AL values and updated the NOR with +/- 20% of the target. This change was made to control the parameter based on lack of

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					osmolality testing on the DS spec.
	0.2µm Bulk Filtration of FB to DS	Capacity (L/m^2)	**ED001:** NOR max = TBD	**ED002:** NOR max = 245	Defined challenge for manufacturing.
		Drug Substance storage temperature (^{o}C)	**ED001:** Held for 14 days at 2-8°C prior to shipment.	**ED002:** Held for 7 days at 2-8°C or 3 days at 15-25°C prior to freeze.	Hold times revised based on updated data.

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14 EDITION HISTORY

Edition	Reason
1.0	New document
2.0	See process changes in Section 13. Revised hold times in Table 3.1. Revised load and product filter parameters in Table 4.1. Added and revised notes in Table 5.1, Table 6.1, and Table 7.1 to provide additional manufacturing guidance. Added density of diluted mAb intermediate in Table 5.1. Increased NOR maximum incubation time for reduction in Table 6.1. Defined NOR min and max molar addition ratio for TCEP in Table 6.1. Defined NOR min and max molar addition ratios for payload in DMSO solution and NAC solution in Table 7.1. Increased NOR maximum incubation time for quench in Table 7.1. Added density of crude conjugate in Table 7.1 Added NOR min and max ranges for process outputs in Table 8.1. Added Strip 1 step in Table 8.1. Revised elution collection criteria in Table 8.1. Revised AL max and NOR max membrane protein load in Table 9.1. Revised targets, ALs, and NORs for concentrations in Table 9.1. Added osmolality target and AL to Table 10.1. Added filter capacity in Table 10.1. Revised Drug Substance hold time before freeze in Table 10.1. Revised buffer information in Table 11.1 and throughout Section 11. New buffer added to Table 11.1 and added buffer recipe in Table 11.9 and Table 11.12.
3.0	Updated Formulated Bulk hold time at room temperature from ≤ 48 hours to ≤ 72 hours. Revised temperature target and min AL values in table 7.1 to ensure min NOR is not exceeded due to temperature fluctuation. Updated Solution and buffer table with reference to Pro-0188988 for buffer recipes and removed table 11.2 to 11.12 from the document. Added buffer hold time column in Table 11.1 for all buffers and added platform language for buffer hold time to align with platform in section 11. Updated table 10.1 with updated value for target osmolality and NOR and removed AL values to conform with guidance, No Osmolality in the Master specification. Updated signature page with current ECMS template. Deleted document header to conform with ECMS template Updated Edition numbers from 001 to 1.00 and from 002 to 2.00 to conform with ECMS template.
4.0	Updated Section 7, Table 7-1: Added a parameter for payload in DMSO addition time to control payload addition per QE-097441.

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Edition	Reason
	Updated Section 8, Table 8-1 CHT Load step: Increased max load to 45g/L from 30 g/L per QE-097441. Updated Section 8, Table 8-1 CHT: Added notes under Elution step to provide information on volume prior to pre-peak and main peak, and expected elution volume. Updated Table 3.1: Changed Hold times for Quenched intermediate from 12hrs to 24 hours for room temperature and 2-8 °C Hold times per QE-097441. Corrected error in process changes table 13-1: changed CHT Start Collection OD from 2.0 to 1.0 to match section 8 table. Removed all alert limits references from the process description per new platform guidance. Updated all references to EQV values.

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